

ApexCloud & FinSight AI Technical Product Manual

Product Version: v3.4 Enterprise | Architecture Specification & SLA

1. System Architecture Overview

ApexCloud Enterprise provides distributed data integration and intelligent agent runtime services. The system leverages a dual-engine architecture combining dense vector retrieval (ChromaDB with BAAI/bge-small-en-v1.5) and sparse inverted index matching (BM25) routed through an ensemble weighting formula (0.6 Dense / 0.4 Sparse).

The agentic runtime is managed by LangGraph with Groq's high-speed Llama-3.3-70B inference engine, maintaining an average token generation latency of less than 1.5 seconds per multi-tool hop.

2. Service Level Agreements (SLA) & Uptime Guarantees

Tier	Availability Guarantee	Target Response Time (P1)	Monthly Credit on Breach
Standard Tier	99.5% Uptime	< 4 Hours	10% of monthly bill
Enterprise Tier	99.9% Uptime	< 1 Hour	25% of monthly bill
Mission-Critical Tier	99.99% Uptime	< 15 Minutes (24/7/365)	50% of monthly bill

3. API Rate Limits & Authentication

Authentication requires an enterprise Bearer token in the Authorization header. Rate limits are enforced at the API Gateway: **Enterprise Tier** accounts are permitted up to **1,200 requests/minute** with bursting up to 2,000 requests. Exceeding thresholds triggers HTTP 429 (Too Many Requests).

4. Text-to-SQL Security Guardrails

The SQL tool strictly executes in SQLite Read-Only mode (?mode=ro). Any SQL statement containing mutating keywords (e.g. DROP, DELETE, INSERT, UPDATE, ALTER) is blocked by the parser before hitting the execution engine.